

CHEMISTRY

TARGET : JEE- Advanced

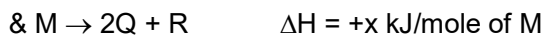
CAPS-17**CHEMICAL KINETICS****SINGLE CHOICE QUESTIONS**

1. For the system $A \xrightleftharpoons[k_{-1}]{k_1} B, A \xrightleftharpoons[k_{-2}]{k_2} C$ which was started with only A, the equilibrium concentration

$[A]_{eq}$ is correctly related to $[B]_{eq}$ and $[C]_{eq}$ as

- (A) $\frac{k_{-1}[B]_{eq} + k_{-2}[C]_{eq}}{(k_1 + k_2)}$ (B) $\frac{k_{-1}[B]_{eq} - k_{-2}[C]_{eq}}{(k_1 - k_2)}$
 (C) $\frac{k_{-1}[B]_{eq} + k_{-2}[C]_{eq}}{(k_1 - k_2)}$ (D) $\frac{k_{-1}[B]_{eq} - k_{-2}[C]_{eq}}{(k_1 - k_2)}$

2. For a hypothetical reaction,



These reactions are carried simultaneously in a reactor such that temperature is not changing.

If rate of disappearance of B is $y \text{ M sec}^{-1}$, then the rate of formation (in M sec^{-1}) of Q is:

- (A) $\frac{2}{3}y$ (B) $\frac{3}{2}y$ (C) $\frac{4}{3}y$ (D) $\frac{3}{4}y$

3. $A(aq.) \rightarrow B(aq.) + C(aq.)$ is a first order reaction.

Time $t \rightarrow \infty$

moles of reagent $n_1 \quad n_2$

Reaction progress is measured with the help of titration of reagent 'R'. If all A, B and C react

with reagent and have 'n' factors [n factor; eq. mass = $\frac{\text{mol. mass}}{n}$] in the ratio of 1 : 2 : 3 with

the reagent, the k in terms of t , n_1 and n_2 is:

- (A) $k = \frac{1}{t} \ln \left(\frac{n_2}{n_2 - n_1} \right)$ (B) $k = \frac{1}{t} \ln \left(\frac{2n_2}{n_2 - n_1} \right)$ (C) $k = \frac{1}{t} \ln \left(\frac{4n_2}{n_2 - n_1} \right)$ (D) $k = \frac{1}{t} \ln \left(\frac{4n_2}{5(n_2 - n_1)} \right)$

4. 65.6 mg of Na_3PO_4 contains 20% P^{32} and 80% P^{31} atoms. Assuming only radio activity of P^{32} , if the half-life period for $\text{P}^{32} = 14.3$ days, the rate of decay of P^{32} is nearly (Take Mol. wt. of $\text{Na}_3\text{PO}_4 = 164$)

- (A) $2.7 \times 10^{15} \text{ dps}$ (B) $8.4 \times 10^{15} \text{ dps}$ (C) $6.7 \times 10^{15} \text{ dps}$ (D) $27 \times 10^{12} \text{ dps}$



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MCQ (One or more than one correct) :

5. Select the correct statements(s):
- (A) When $T \rightarrow \infty$ or $E_a \rightarrow 0$ then $k = A$.
- (B) A positive catalyst can change ΔH of the reaction.
- (C) A mixture of reactants may be thermodynamically unstable but kinetically stable.
- (D) A negative catalyst increases the activation energy of the reaction.
6. A reaction takes place in three steps. The rate constant of the three steps are K_1 , K_2 and K_3 respectively. The overall rate constant $K = \frac{K_1 K_3}{K_2}$.

The energy of activation for the three steps are 40, 30 and 20 KJ respectively. Therefore :

- (A) Overall energy of activation is 10 KJ
- (B) Overall energy of activation is 30 KJ
- (C) The reaction mechanism is $2A \xrightleftharpoons[K_2]{K_1} A^* + A; A^* \xrightarrow[\text{rds}]{K_3} \text{product}$ and overall order is one.
- (D) The reaction mechanism is $A \xrightarrow{K_1} B; B \xrightarrow{K_2} C; C \xrightarrow{K_3} \text{product}$ and overall order is one
7. The rate expression for the reaction :

$\text{NH}_4\text{CNO} \rightleftharpoons \text{NH}_2\text{CONH}_2$ can be derived from the mechanism :

- i. $\text{NH}_4\text{CNO} \xrightleftharpoons[k_2]{k_1} \text{NH}_4\text{NCO}$ (Fast)
- ii. $\text{NH}_4\text{NCO} \xrightarrow{k_3} \text{NH}_3 + \text{HNCO}$ (Fast)
- iii. $\text{NH}_3 + \text{HNCO} \xrightarrow{k_4} \text{NH}_2\text{CONH}_2$ (Slow)

Which of the following statement(s) is/are correct about the rate expression ?

- (A) $\frac{d_{(\text{urea})}}{dt} = \frac{k_1 k_3}{k_2} [\text{NH}_4\text{NCO}]$
- (B) $\frac{d_{(\text{urea})}}{dt} = \frac{k_1 k_3}{k_2 k_4} [\text{NH}_4\text{CNO}]$
- (C) $\frac{d_{(\text{urea})}}{dt} = k [\text{NH}_4\text{NCO}]$
- (D) $\frac{d_{(\text{urea})}}{dt} = \frac{k_1 \times k_2}{k_3 \times k_4} [\text{NH}_4\text{NCO}]$

Comprehension Type Question:

Study the two photochemical reactions and answer the question at the end.

For the overall reaction between A and B to yield C and D, two mechanisms are proposed :

- I. $A + B \longrightarrow AB^* \longrightarrow C + D$, $k_1' = 1 \times 10^{-5} \text{ M}^{-1} \text{ s}^{-1}$
- II. $A \longrightarrow A^* \longrightarrow E$, $k_1 = 1 \times 10^{-4} \text{ s}^{-1}$
- $E + B \longrightarrow C + D$, $k_2 = 1 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$

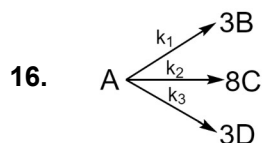
(species with * are short-lived)

8. Rate according to mechanism I when concentration of each reactant is 0.1 M will be :
- (A) $1 \times 10^{-7} \text{ Ms}^{-1}$ (B) $1 \times 10^{-6} \text{ Ms}^{-1}$ (C) $1 \times 10^{-5} \text{ Ms}^{-1}$ (D) $1 \times 10^{-4} \text{ Ms}^{-1}$

9. Rate according to mechanism II when concentration of each reactant is 1 M will be:
 (A) $1 \times 10^{-4} \text{ Ms}^{-1}$ (B) $1 \times 10^{10} \text{ Ms}^{-1}$ (C) $1 \times 10^{-6} \text{ Ms}^{-1}$ (D) $1 \times 10^{-10} \text{ Ms}^{-1}$
10. At what concentration of B, rates of two mechanism are equal :
 (A) 1 M (B) 5 M (C) 7 M (D) 10 M

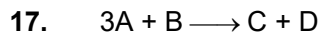
Numerical based Questions :

11. Decomposition of H_2O_2 is a first order reaction. A solution of H_2O_2 labelled as 20 volumes was left open. Due to this, some H_2O_2 decomposed. To determine the new volume strength after 6 hours, 10 mL of this solution was diluted to 100 mL. 10 mL of this diluted solution was titrated against 25 mL of 0.025 M KMnO_4 solution under acidic conditions. Calculate the rate constant for decomposition of H_2O_2 .
12. Potassium having atomic mass = 39.1 u contains 93.10 atom % ^{39}K , having atomic mass 38.96371 u; 0.0118 atom % ^{40}K , which has mass of 40.0 u and is radioactive with $t_{1/2} = 1.3 \times 10^9 \text{ yr}$, and 6.88 atom % ^{41}K having a mass of 40.96184 u. Calculate the specific activity of naturally occurring potassium.
13. A certain reactant B^{n+} is getting converted to $\text{B}^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing reagent which only reacts with B^{n+} and $\text{B}^{(n+4)+}$. In this process, it converts B^{n+} to $\text{B}^{(n-2)+}$ and $\text{B}^{(n+4)+}$ to $\text{B}^{(n-1)+}$. At $t = 0$, the volume of the reagent consumed is 25 mL and at $t = 10 \text{ min}$, the volume used up is 32 mL. Calculate the rate constant of the conversion of B^{n+} to $\text{B}^{(n+4)+}$ assuming it to be a first order reaction.
14. For a reversible first-order reaction $\text{A} \xrightleftharpoons[k_2]{k_1} \text{B}$
 $k_1 = 10^{-2} \text{ s}^{-1}$ and $[\text{B}]_{\text{eq}} / [\text{A}]_{\text{eq}} = 4$. If $[\text{A}]_0 = 0.01 \text{ mole L}^{-1}$ and $[\text{B}]_0 = 0$, what will be the concentration of B after 30 s?
15. A small quantity of solution containing ^{24}Na radionuclide (half-life 15 hours) of activity 1.0 microcurie is injected into the blood of a person. A sample of the blood of volume 1 cm^3 taken after 5 hours shows an activity of 296 disintegrations per minute. Determine the total volume of blood in the body of the person. Assume that the radioactive solution mixes uniformly in the blood of the person. (1 Curie = 3.7×10^{10} disintegrations per second)

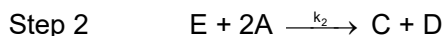


At time $t = 0$, initial mole of A is 1.

Overall half-life of the reaction is 15 days. Then calculate the number of mole of C after 45 days if the ratio of $k_1 : k_2 : k_3$ is 4 : 2 : 1



Mechanism of given reaction is :



What is the total order of reaction considering steady state approximation.

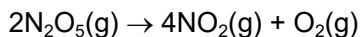
18. A reaction take place at 300K. When catalyst is added rate of reaction increases. Calculate the change in temp (in °C) which would have same effect on rate of reaction as that of catalyst. (Assume catalyst change the activation energy by 20%).

SUBJECTIVE ANSWER TYPE

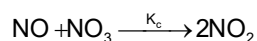
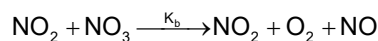
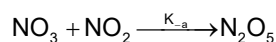
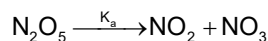
19. The conversion of A into B is an auto-catalytic reaction $A \rightarrow B$ where B catalyzes the reaction. The rate equation is $-dx/dt = kxy$ where x and y are concentrations of A and B at time t. Integrate this equation for initial concentration x_0 and y_0 for A and B. Show that:

$$kt = \frac{2.303}{x_0 + y_0} \log \frac{x_0 y}{xy_0}$$

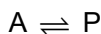
20. Derive an expression for the rate of reaction:



With the help of following mechanism



21. For the reaction

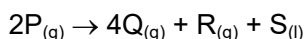


Following data is produced

Time/Hr.	0	1	2	3	4	∞
%A	100	72.5	56.8	45.6	39.5	30

Find k_1 , k_{-1} and K_{eq} .

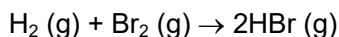
22. The decomposition of a compound P, at temperature T according to the equation



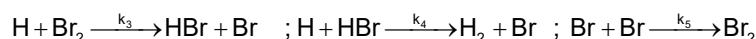
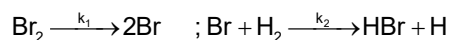
is the first order reaction. After 30 minutes from the start of decomposition in a closed vessel, the total pressure developed is found to be 317 mm Hg and after a long period of time the total pressure observed to be 617 mm Hg. Calculate the total pressure of the vessel after 75 minute, if volume of liquid S is supposed to be negligible. Also calculate the time fraction $t_{7/8}$. Given: Vapour pressure of $S_{(l)}$ at temperature T = 32.5 mm Hg.

23. The acid catalysed hydrolysis of an organic compound A at 30°C has a time for half change of 100 minutes, when carried out in a buffer solution at pH 5 and of 10 minutes, when carried out at pH 4. Both times of half change are independent of the initial concentration of A. If the rate of reaction is given by $\frac{-d[A]}{dt} = k[A]^a [H^+]^b$, what are the values of a and b?

24. Derive an expression for the rate of reaction:



With the help of following mechanism



25. For the reaction $A \rightarrow B$, the rate law expression is $-\frac{d[A]}{dt} = K[A]^{1/2}$. If initial concentration of A is A_0 . Calculate :
 (a) integrated form of the rate law expression
 (b) nature of plot of $[A]^{1/2}$ vs time.
 (c) half life period.
26. (a) Let there be as first-order reaction of the type, $A \rightarrow B + C$. Let us assume that all three species are gases. We are required to calculate the value of rate constant based on the following data.

Time	0	T	∞
Partial pressure of A	P_0	P_t	—

- (b) Let there be a first order reaction, $A \rightarrow B + C$. Let us assume all three are gases. We are required to calculate the value of rate constant based on the following data

Time	0	t	∞
Total pressure	P_0	P_t	—

Calculate the expression of rate constant.

- (c) $A(g) \rightarrow B(g) + C(g)$

Time	0	t	∞
Total pressure of B + C	—	P_t	P_∞

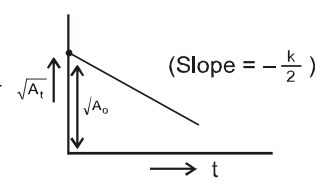
Calculate the expression of rate constant.

- (d) $A(g) \rightarrow B(g) + C(s)$

Time	0	t	∞
Total pressure of B	—	P_t	P_∞

Calculate the expression of rate constant.

ANSWER KEY CAPS-17

1. (A) 2. (C) 3. (D) 4. (D) 5. (ACD) 6. (BC) 7. (AC)
 8. (A) 9. (A) 10. (D)
 11. ($k=0.022 \text{ hr}^{-1}$) 12. ($30.7242 \text{ dis g}^{-1}\text{s}^{-1}$) 13. (0.0207 min^{-1})
 14. (0.0025 M) 15. (6 litre) 16. (2.00)
 17. (2) 18. (75) 19. $\frac{2.303}{(x_0 + y_0)} \log \frac{yx_0}{xy_0}$
 20. $r = \frac{-1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = \frac{k_a k_b [\text{N}_2\text{O}_5]}{k_{-a} + 2k_b}$ 21. $9.7 \times 10^{-5} \text{ sec}^{-1}$ 22. $t = 399.89 \text{ min.}$
 23. ($a = 1, b = 2$) 24. $\frac{2k_2 (k_1 / k_5)^{1/2} [\text{Br}_2]^{1/2} [\text{H}_2]}{1 + (k_4 / k_3) [\text{HBr}] / [\text{Br}_2]}$
 25. (a) $\sqrt{A_t} = \sqrt{A_0} - \frac{kt}{2}$ (b) $\sqrt{A_t} = \left(\frac{-k}{2}\right)t + \sqrt{A_0}$

 (c) $t_{\frac{1}{2}} = \frac{2}{k} \left[\frac{\sqrt{2}-1}{\sqrt{2}} \right] \sqrt{A_0} = \frac{\sqrt{2}(\sqrt{2}-1)\sqrt{A_0}}{k} = t_{1/2}$
 26. (a) $k = \frac{1}{t} \ln \left(\frac{P_0}{P_t} \right)$ (b) $k = \frac{1}{t} \ln \left(\frac{P_0}{2P_0 - P_t} \right)$
 (c) $K = \frac{1}{t} \ln \left(\frac{P_\infty}{P_\infty - P_t} \right)$ (d) $k = \frac{1}{t} \ln \left[\frac{P_\infty}{P_\infty - P_t} \right]$